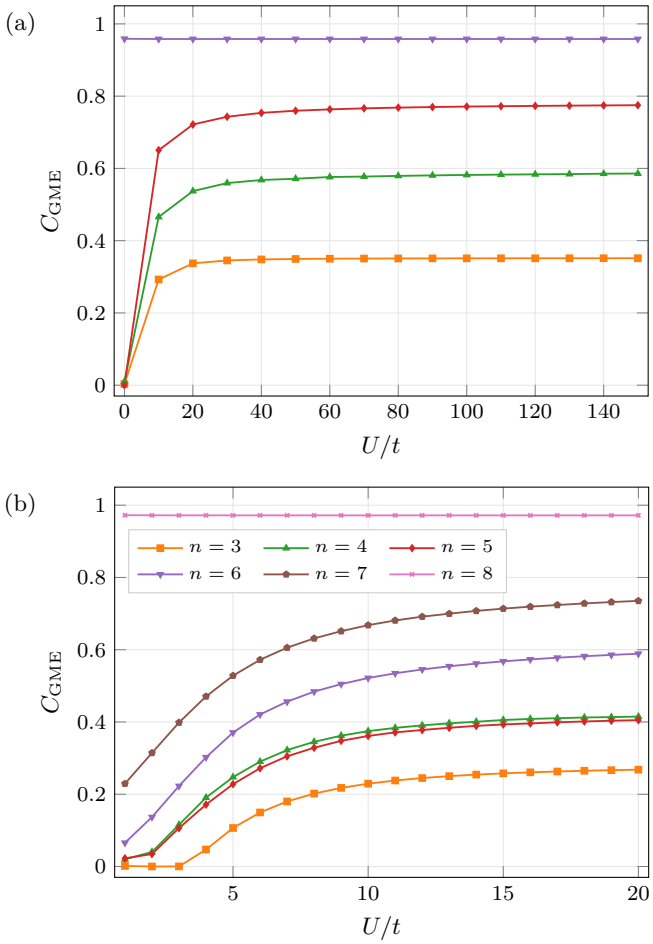

FIG. 1. Spin entanglement in the ground state of the Hubbard ring as a function of U/t, for different values of the spin number n, with $K = L$ and $V = 0$: (a) $L = 6$, (b) $L = 8$.

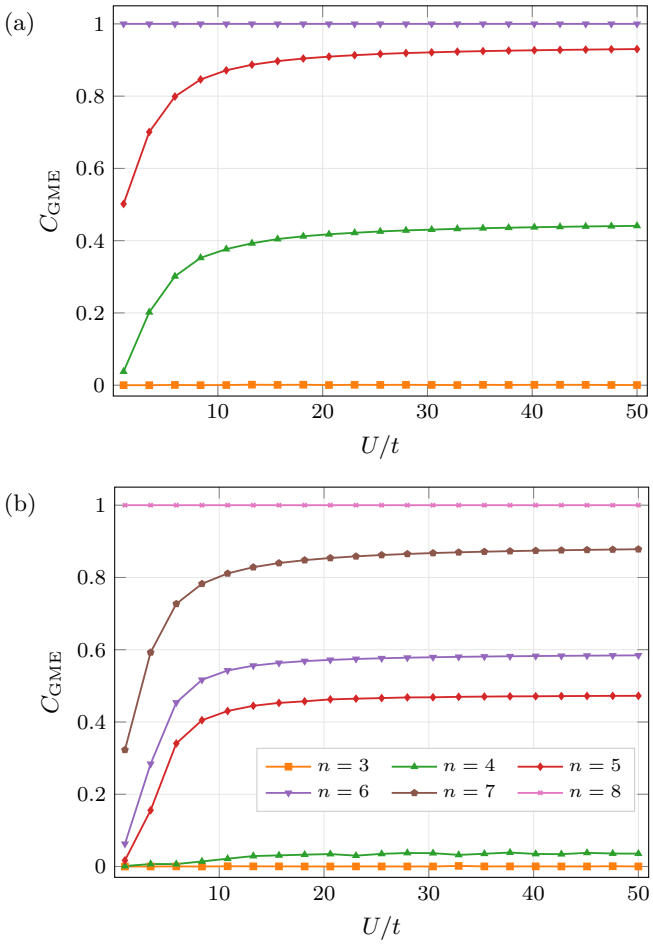

FIG. 2. Spin entanglement in the first excited singlet state of the Hubbard ring as a function of U/t, for different values of the spin number n, with $K = L$ and $V = 0$: (a) $L = 6$, (b) $L = 8$.

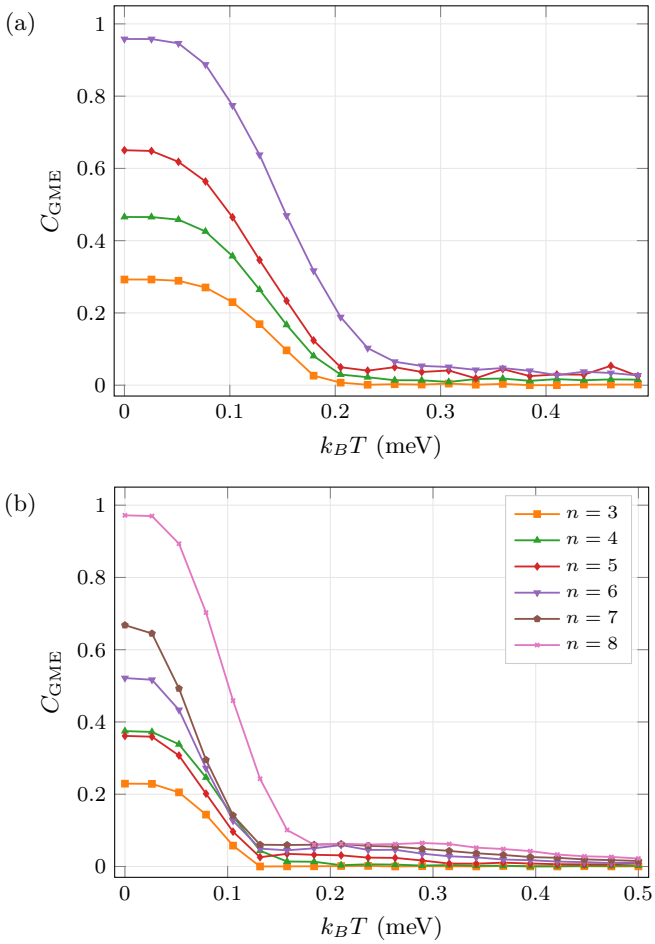

FIG. 3. Temperature dependence of the spin entanglement of the Hubbard ring at $U = 10$, $t = 1$: (a) $L = 6$, (b) $L = 8$.